

ARTIFICIAL INTELLIGENCE – ASSESSMENT 1 REPORT

Course Code: CSA1704 – Artificial Intelligence

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Introduction

Artificial Intelligence (AI) is a rapidly growing field of computer science that focuses on developing intelligent systems capable of performing tasks that normally require human intelligence. These tasks include learning, reasoning, problem-solving, planning, decision-making, perception, and natural language processing.

AI uses various techniques such as search algorithms, intelligent agents, and optimization methods to solve complex problems efficiently. In this assessment, five fundamental AI concepts are studied and

implemented using Python. These problems demonstrate state-space search, intelligent agents, constraint satisfaction, and graph search algorithms, providing practical knowledge of Artificial Intelligence concepts.

Objectives

The main objectives of this assessment are:

- To understand the fundamental concepts of Artificial Intelligence.
- To study state-space search techniques and problem-solving methods.
- To design and implement intelligent agents.
- To solve constraint satisfaction problems using AI techniques.
- To implement graph search algorithms using Python.
- To gain practical knowledge through Python programming and algorithm implementation.